MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE SET: M32

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of an electrical assembly entitled as Semiconductor Device Set, M32 used on the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8808362	- Container, Packing, Ammunition, for Domestic (CONUS) Shipments of Special Weapons Items. Packing and Marking.
8822044	- Ammunition Packing Carton for Semiconductor Device Set M32.
8826411	- Warhead Section, Atomic: M27.
8826440	- Semiconductor Device Set M32.
8826441	- Schematic Diagram Semiconductor Device Set M32.
EL-8842565	- Equipment Lists.

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PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instruction for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The semiconductor device set shall comply with all requirements specified on the drawing (dwg) 8826440, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SMUPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Circuit resistance before and after operation. The resistance of each circuit shall not exceed the value stated for the respective circuit in Table I. The test shall be conducted as specified in 4.4.1.

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Table I

Continuity
(For Circuit reference see dwg 8826411)

From Terminal P1	To Terminal(s) P1	Maximum Resistance (unless stated) in Ohms Before Operation -65° ± 5°F +80° ± 20°F	Before Operation +160° ± 5°F	After Operation -65° ± 5°F +80° ± 5°F	After Operation +160° ± 5°F	Test Code No.
1	2,11,29,30	0.041	0.050	0.041	0.050	
3	21,22	0.041	0.050	0.041	0.050	
6	25	0.041	0.050	0.041	0.050	
8	27	0.041	0.050	0.041	0.050	
7	8	0.090	0.110	--	--	030001
9	26	0.090	0.110	--	--	030002
13	14	0.090	0.110	--	--	030003
31	33	0.090	0.110	--	--	030004
15	33	0.041	0.050	0.041	0.050	
28	9	0.041	0.050	0.041	0.050	
17	36,35	0.041	0.050	0.041	0.050	
18	19,37	0.041	0.050	0.041	0.050	
14	32	0.041	0.050	0.041	0.050	
36	2	2.4 min 3.6 max	2.4 min 3.6 max	2.4 min 3.6 max	2.4 min 3.6 max	
37	2	2.4 min 3.6 max	2.4 min 3.6 max	2.4 min 3.6 max	2.4 min 3.6 max	030005 030006

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3.7 Insulation resistance before and after operation. The insulation resistance of each terminal to all other terminals listed in Table II shall be greater than 20 megohms when tested at 500 ± 50 volts direct current (vdc). Testing shall be as specified in 4.4.2.

Table II

Insulation resistance
(For circuit reference see dwg 8826441)

Terminal Before Operation	Terminal After Operation
P1 - 1	P1 - 1
6	6
7	7
9	8
13	9
31	13
Unused terminals 4,5,10	14
12,16,20,23,24 and 34	26
Note 1	
Ground-this will be	31
the test fixture ground	33
- Note 2	
	Unused terminals 4,5,10,12, 16,20,23,24, and 34 Note 1 Ground-this will be the test fixture ground-Note 2

Note 1 Unused terminal may be connected together and tested as one terminal.
Note 2 Connector shell shall be at ground potential.

3.8 Leakage current. The leakage current from terminal 21 to terminal 1 shall not be more than 1.0 milliampere when tested at 75 ± 5 vdc. Testing shall be as specified in 4.4.3.

From Pin	To Pin	Actual Current Amperes	Specified Current Amperes	Test Code No.
21	1	X10^{-3}	1.0 X 10^{-3} max	050001

3.9 Semiconductor operation. Each item shall be capable of meeting the requirements and conditions specified in Table III when tested as specified in 4.4.4.

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Table III

Test	Anode Power Source 30 ± 2 vdc Positive	Anode Power Source 30 ± 2 vdc Negative	D.C.Volts	Signal Source*** Positive	Signal Source*** Negative	Results Anode Current	Results Anode Voltage	Test Code No.
Non-operate 1	P1-3, P1-22 in parallel	P1-1, P1-11 in parallel	0.4 -0.02	P-18, P1-19 in parallel	P1-29 P1-30 in parallel	0.001 amps	--	060001
Non-operate 2	P1-3, P1-22 in parallel	P1-1, P1-11 in parallel	0.4 -0.02	P1-35, P1-36 in parallel	P1-29 P1-30 in parallel	0.001 amps	--	060002
Operate 2	P1-3, P1-22 in parallel	P1-1, P1-11 in parallel	2.0 +0.2	P1-35, P1-36 in parallel	P1-29, P1-30 in parallel	** 45 ± 5 amp	3.2 volts max	060003
Operate 1	P1-3, P1-22 in parallel	P1-1, P1-11 in parallel	2.0 +0.2	P1-18, P1-19 in parallel	P1-29, P1-30 in parallel	** 45 ± 5 amp	3.2 volts max	060004
Operate 1 & 2	P1-3, P1-22 in parallel	P1-1, P1-11 in parallel	6.0 ±0.5	P1-18, P1-19 in parallel	P1-29, P1-30 in parallel	** 45 ± 5 amp	2.7 volts max	060005

**Load current duration 30 milliseconds maximum.
***The signal source shall supply with voltage specified when connected between the pins indicated.

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3.10 Fuse resistance. The voltage across the terminals listed in Table IV, measured over a time interval from 0.04 ± 0.02 seconds to 2 seconds minimum after application of a current of 1.5 ± 0.10 amperes, shall increase a minimum of 7.5 millivolts when tested as specified in 4.4.5 at a temperature of +80 ± 20°F.

Table IV

Fuse	From Terminal	To Terminal	Minimum Voltage Rise
F1	7	8,27	7.5 millivolt
F2	26	9,28	7.5 millivolt
F3	13	14,32	7.5 millivolt
F4	31	15,33	7.5 millivolt

3.11 Fuse operation. After F1, F2, F3, and F4 have operated, the assembly shall meet the requirements of 3.7.

3.11.1 Fuses F1 and F2. Fuses F1 and F2 shall each be capable of conducting a current of 8.0 + 0, -0.25 amperes for a period of at least 75 milliseconds and shall open the circuit within 300 milliseconds. Testing shall be as specified in 4.4.6.1.

3.11.2 Fuses F3 and F4. Fuses F3 and F4 shall each be capable of conducting a current of 30 + 0, -0.5 amperes for a period of at least 1 millisecond and shall open the circuit within 20 milliseconds. Testing shall be as specified in 4.4.6.2.

3.12 Environmental testing. The assembly shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The assembly shall also comply with applicable paragraphs in this section before and after each environmental test. Testing shall be as specified in 4.4.7. Tests shall be conducted in the order specified.

3.12.1 Transportation vibration. The assembly shall be vibrated at the frequencies, gravitational units (g), acceleration, temperatures, and for the times specified in Table V without evidence of damage such as cracks, chips, separations, loose parts, etc and shall comply with 3.6 through 3.10 after being tested as specified in 4.4.7.1.

Table V

Frequency Range Hertz(Hz)	Double Displacement or Peak g Level (± 5% tol)	Cycle Time to Cover Frequency Range	Test Time Per Axis in Minutes at Given Temperature (-10% tol) +180 + 20°F	+160 + 5°F	-65 + 5°F
10-20-10	2 g	3 ± 0.5 min	15	10	10
20-50-20	0.036 inch	3 ± 0.5 min	15	10	10
50-500-50	5 g	3 ± 0.5 min	15	10	10

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3.12.2 Flight vibration. The assembly shall withstand subjection to vibration over a frequency range of 75 to 1000 Hz at a peak level of 5 g's ± 5 percent and at a temperature of 80 degrees Fahrenheit (°F) plus or minus (±) 20°F. The vibration shall be applied from a low to a high frequency and returned to low frequency over a period of 3.0 ± 0.3 minutes in each of three mutually perpendicular axes, one of which is in the direction of forward motion (see Figure 1). Semiconductor operation and fuse continuity shall not be interrupted for longer than 100 microseconds during this test. Testing shall be as specified in 4.4.7.2. Upon completion of this test the assembly shall comply with 3.6 through 3.10.

3.12.3 Acceleration. The assembly shall withstand, without malfunction, acceleration of 50 ± 5 g's for 5 seconds in the direction of forward motion (see Figure 1) and 18 ± 1 g's for 15 seconds perpendicular to the direction of forward motion. The assembly shall comply with 3.6 through 3.10 after testing in accordance with 4.4.7.3.

3.12.4 Transportation shock. The assembly shall be capable of withstanding shocks of 75 ± 8 g's peak amplitude with a half amplitude duration of 9 ± 3 milliseconds. The assembly shall comply with 3.6 through 3.10 after being tested as specified in 4.4.7.4.

3.12.5 Impact shock. The assembly shall be capable of withstanding a shock of 150 ± 15 g's peak with a half amplitude duration of 4 ± 1 milliseconds applied once in the axial direction opposite that of forward motion. The assembly shall meet fuse continuity requirements of 3.6 while being tested as specified in 4.4.7.5.

3.12.6 High temperature. The assembly shall withstand subjection to +160 ± 5°F without damage and shall comply with 3.6 through 3.9 and 3.11 while at 160 ± 5°F. The insulation resistance requirements of 3.7 shall be modified to 1 megohm minimum for this test. Testing shall be as specified in 4.4.7.6.

3.12.7 Low temperature. The assembly shall withstand subjection to -80 ± 5°F without damage and shall comply with 3.6 through 3.9 and 3.11 while at -65 ± 5°F when tested as specified in 4.4.7.7.

3.13 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.14 Test codes (see 3.5). The contractor shall provide data feedback on punch cards in conformance with Quality Assurance Instructions QAI-PA-1. The data required to be coded, and the code numbers, are specified in the applicable paragraphs in this section.

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3.15 Workmanship.

3.15.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.15.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.15.3 Burrs. Parts shall be free from burrs.

3.15.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.15.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.15.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 8 complete assemblies (see dwg 8826440) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VI. Tests shall be conducted in order specified in Table VI.

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Table VI
First article inspection and tests
(4 units per group)

Inspection or Test	Group I	Group II	Dest	Non Dest
1. Classification of characteristics (see 4.3.2.1)	X	X		X
2. Circuit resistance (see 3.6)	X	X		X
3. Insulation resistance (see 3.7)	X	X		X
*4. Leakage current (see 3.8)	X	X		X
*5. Semiconductor operation (see 3.9)	X	X		X
*6. Fuse resistance (see 3.10)	X	X		X
7. Fuse operation (see 3.11)	X	X	X	
8. Transportation vibration (see 3.12.1)	X	X	X	
9. Acceleration (see 3.12.3)	X	X		X
10. Transportation shock (see 3.12.4)	X	X	X	
11. Impact shock (see 3.12.5)	X	X		X
12. Flight vibration (see 3.12.2)	X	X		X
13. High temperature (see 3.12.6)	X			X
14. Low temperature (see 3.12.7)		X		X

*Testing sequence optional

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

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4.3.1 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VII and Table VIII), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table VII
Major characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VIII
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Semiconductor device set, M32, prior to potting (see dwg 8826440).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper soldering (see 3.13)	Visual
	102. Evidence of poor workmanship (see 3.15)	Visual
	103. Any component missing, damaged or loose	Visual/Manual
	104. Improper wiring (wires in tension, excess wire lengths, inadequate or damaged insulation	Visual/Manual
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.2 Semiconductor device set (see dwg 8826440).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper location of mounting holes and connector	Gage
	102. Deformation of connector	Visual

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103. Inadequate potting.....Visual
104. Any component missing, incorrect or incorrectly assembled.....Visual

Minor:
201. Information on identification plate incorrect.....Visual
202. Serialization missing or improper (see 3.1).....Visual

4.3.2.1.3 Carton, packing, ammunition, for semiconductor device set, M32 (see dwg 8822044).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Excess air in bag	Visual
102.	Packing material visibly wet	Visual
103.	Insufficient cushioning material	Visual/Manual
Minor:		
201.	Marking incorrect or missing	Visual

4.3.2.1.4 Container, packing (see dwg 8808362).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect or insufficient cushioning material	Visual
102.	Marking missing, incorrect or mislocated	Visual
103.	Packing material visibly wet	Visual
104.	Evidence of poor workmanship (see 3.15)	Visual
105.	Contents loose	Visual
Minor:	None defined.	

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Circuit resistance	Major
b. Insulation resistance	Major
c. Leakage current	Major
d. Semiconductor operation	Major

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e.	Fuse resistance	Major
f.	Fuse operation	Major
g.	Transportation vibration	Major
h.	High temperature	Major
i.	Low temperature	Major

Of these tests, (f) and (g) are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Circuit resistance, insulation resistance, leakage current, semiconductor operation and fuse resistance. Each semiconductor device set in the lot shall be tested in accordance with 4.4.1 through 4.4.5. If any item fails to comply with 3.6 through 3.9, the item shall be rejected and removed from the lot.

4.3.3.2 Fuse operation, transportation vibration, high temperature and low temperature. A sample shall be selected at random in accordance with Table IX. The sample shall be tested in accordance with 4.4.6, 4.4.7.1, 4.4.7.6 and 4.4.7.7. If any item fails to comply with 3.11, 3.12.1, 3.12.6 and 3.12.7, the lot shall be rejected.

Table IX
Sample sizes for destructive testing

Number Required to be Delivered	Sample Size for No. Failures
3-500	3*
501-800	6*
801-1300	9*

Lot Size Required = Number Required to be Delivered + Sample Size

*All samples shall be subjected to transportation vibration then the samples shall be devided equally among fuse operation, high temperature and low temperature.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8842565). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).

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b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance. The circuit resistance shall be measured by means of a suitable resistance measuring instrument which has been connected to the specified terminals. The current shall be limited to five milliamperes.

4.4.2 Insulation resistance. An insulation resistance tester capable of supplying the required voltage and with which resistance in megohms can be obtained, shall be used to measure the resistance. The current shall be limited to a maximum of five milliamperes to protect human life.

4.4.3 Leakage current. A voltage of 75 ± 5 vdc shall be applied from pin 21 (positive) to pin 1 (negative) and the leakage current shall be measured on an instrument capable of measuring 1.0 milliamperes.

4.4.4 Semiconductor operation. Apply a potential of 30 ± 2 vdc in accordance with Table III and prior to signal voltage application. Current shall be limited to 50 amperes maximum and the test time shall not exceed 30 milliseconds for "operate" tests. Signal voltage shall be applied as specified in Table III. Anode currents shall be measured with instruments capable of measuring 1.0 milliamperes and 50 amperes. Diode voltage drops shall be measured with an instrument capable of measuring a change from 33 volts to 2.7 volts or less within 30 milliseconds. POwer shall be interrupted between tests in Table III.

4.4.5 Fuse resistance. The resistance change of each fuse of Table IV shall be measured by application of a constant current of 1.5 ± 0.10 amperes for a minimum of 2 seconds. The voltage rise which occurs 0.04 ± 0.02 seconds and 2 seconds after the current is applied shall be measured at the fuse terminals with instrumentation capable of detecting a 7.5 millivolt change within the time limits specified.

4.4.6 Fuse operation. After F1, F2, F3 and F4 have operated, the assembly shall be tested as specified in 4.4.2.

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4.4.6.1 Fuses F1 and F2 operation. Apply a current of 8.0 + 0, -0.25 amperes dc through pin 7 to parallel connected pins 8 and 27. a suitable time interval device shall be used to measure the interval from current initiation to current interruption. Repeat the above procedure for current through pin 26 to parallel connected pins 9 and 28.

4.4.6.2 Fuses F3 and F4 operation. apply a current of 30 + 0, -0.25 amperes dc thru pin 13 to parallel connected pins 14 and 32. A suitable time interval device shall be used to measure the interval from the current initiation and one millisecond and from current initiation and current interruption or 20 milliseconds. Repeat the above procedure for a current thru pin 31 to parallel connected pins 15 and 33.

4.4.7 Environmental testing (see 3.12).

4.4.7.1 Transportation vibration. The assembly shall be rigidly mounted to a fixture capable of being supported readily in each of three mutually perpendicular axes, with the fixture mounted on a suitable vibration producing apparatus. The assembly shall be vibrated at the frequencies, amplitudes, temperatures, and for the time specified in Table V in each of three mutually perpendicular axes. Upon completion of this test the assembly shall be tested as specified in 4.4.1 through 4.4.5.

4.4.7.2 Flight vibration. The assembly shall be rigidly mounted to a fixture capable of being supported readily in each of three mutually perpendicular axes with one of these axes along the direction of forward motion (see Figure 1). The assembly shall be vibrated over a frequency range of 75 to 1000 Hz at a peak level of 5 g's ± 5 percent and at a temperature of +80° ± 20°F. The vibration shall be applied from a low to a high frequency and returned to a low frequency over a period of 3 ± 0.3 minutes in each of three mutually perpendicular axes. Fuse continuity and anode current of diodes shall be monitored throughout this test. All monitoring shall be done with instruments capable of detecting a 100 microsecond interruption. A current of 0.5 ± 0.2 amperes shall be used to monitor the continuity of each fuse. A current of 10 + 0, -2 amperes total shall be used to monitor the anode current of all diodes. All diodes shall be gated by applying 6.0 ± 5 vdc between P-18 and P-19 connected inparallel and P1-29, 30 immediately prior to the application of anode current and the removed. No gate signal shall be applied during vibration. Upon completion of this test the assembly shall be tested as specified in 4.4.1 through 4.4.5.

4.4.7.3 Acceleration. Mount the assembly on the centrifuge so that the forward direction of the assembly is oriented to extent in the radial direction of the centrifuge. Bring the centrifuge up to the rotantional speed required to produce an acceleration of 50 ± 5 g's in the forward direction of the assembly. Subject the assembly to 50 ± 5 g's in the forward direction for a period of 5 seconds. Mount the assembly on the

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centrifuge so that the forward direction of the assembly extends in a direction prependicular to the radial direction of the centrifuge. Bring the centrifuge up to rotational speed required to produce 18 ± 1 g's in the direction perpendicular to the forward direction of the assembly. Subject the assembly to 18 ± 1 g's in this direction for a period of 15 seconds. Following this test the assembly shall be tested as specified in 4.4.1 through 4.4.5.

4.4.7.4 Transportation shock. The assembly shall be rigidly mounted to a fixture capable of being supported readily in each of three mutually perpendicular axes. The assembly shall be subjected to shocks of 75 ± 8 g's, with a half amplitude duration of 9 ± 3 milliseconds, three times in each of three mutually perpendicular axes. Following this test the assembly shall be tested as specified in 4.4.1 through 4.4.5.

4.4.7.5 Impact shock. The assembly shall be rigidly mounted to a fixture capable of being supported in the forward axial direction. The assembly shall be subjected to a shock of 150 ± 15 g's, with a half amplitude duration of 4 ± 1 milliseconds in the direction opposite to that of forward motion. Monitor the continuity of each fuse by causing a 0.5 ± 0.2 ampere current to be conducted during this test as specified in 4.4.1. Following this test the assembly shall be tested as specified in 4.4.1 through 4.4.5.

4.4.7.6 High temperature. The assembly shall be placed in the test chamber and the temperature of the chamber raised to +160° ± 5°F, with a relative humidity of not more than 20 ± 10 percent. The assembly shall soak at the temperature of +160° ± 5°F for a minimum of 4 hours. While at this temperature (+160° ± 5°F) the assembly shall be tested as specified in 4.4.1 through 4.4.4 and 4.4.6.

4.4.7.7 Low temperature. The assembly shall be placed in a test chamber and the temperature of the chamber lowered to -80° ± 5°F. The assembly shall soak at the temperature of -80° ± 5°F for a minimum of 4 hours. The chamber temperature shall then be raised to -65° ± 5°F and maintained for an additional 1-hour period or until stabilization (see 6.4) is reached, whichever is longer. While at this temperature (-65° ± 5°F), the assembly shall be tested as specified in 4.4.1 through 4.4.4 and 4.4.6.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The assembly shall be packaged in accordance with dwg 8822044 and 8808362 using the specified moisture barrier.

5.1.2 Level C. The assembly shall be packaged in accordance with dwg 8822044 and 8808362 except that the moisture barrier may be omitted.

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5.2 Packing.
5.2.1 Level A and C. The assembly shall be packed in accordance with dwg 8822044 and 8808362.
5.3 Marking. Marking shall be as specified in accordance with dwg 8808362.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:
6.1.1 Procurement requirements.
a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
6.1.2 Contract data requirements (see 3.5). Data specified in 3.6, 3.8, 3.9 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.
6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.
6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.
6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peak (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.
6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1684A dated 24 January 1967.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. NUOR-A022

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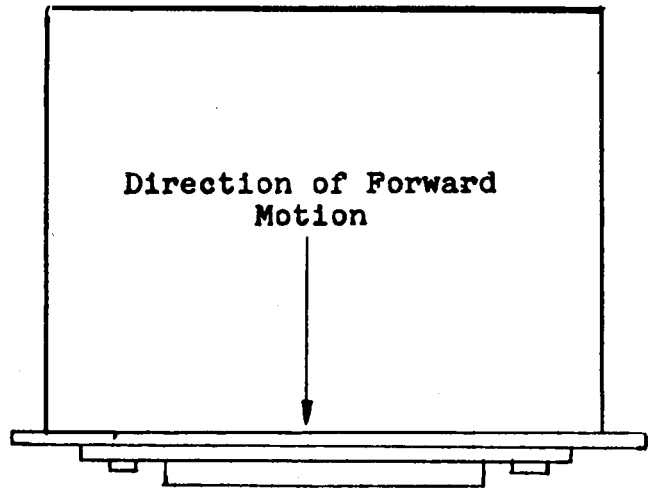


Figure 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
SEMICONDUCTOR DEVICE SET: M32

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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PENALTY FOR PRIVATE USE $300

COMMANDER
Picatinny Arsenal
ATTN: SARPA-QA-N-Q
Dover, New Jersey 07801

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ENGINEERING RELEASE RECORD SARPA-QA-N-Q

1. ERR NO.	2. DATE	3. SHEET OF SHEETS
7400884-2	09/30/74	1 of 4

4. BASELINE ESTABLISHED OR CHANGED: [ ] FUNCTIONAL [ ] ALLOCATED [X] PRODUCT

5. TYPE OF RELEASE: [ ] INITIAL [X] CHANGE

6. ECP NO. _____ DATE APPROVED _____

7. FUNCTIONAL ASSEMBLY NOMENCLATURE: Honest John Warhead Section

8. SYSTEM/SR CONFIGURATION ITEM NOMENCLATURE: Honest John Warhead Section

9. REMARKS/MISCELLANEOUS

10. DATA RELEASED OR REVISED

a CODE ID	b TYPE	c NUMBER	d SH OF SHEETS	e LTR	g DATE	h RELEASE SR	h RELEASE KAA	i CHANGE CN	i CHANGE CAN	j OTHER
		MIL-F-48307	17	-	09/30/74	X				
		PA-PD-1680		-	08/01/61				X	
		MIL-T-48308	24	-	09/30/74	X				
		PA-PD-1683		-	09/06/61				X	
		Amend 1			09/19/61				X	
		MIL-R-48309	16	-	09/30/74	X				
		PA-PD-1689		-	11/06/61				X	
		MIL-S-48310	20	-	09/30/74	X				
		PA-PD-1684		A	01/24/67				X	
		MIL-W-48311	18	-	09/30/74	X				
		PA-PD-1674		A	01/24/67				X	
		MIL-T-48312	16	-	09/30/74	X				
		PA-PD-1671		-	11/07/61				X	
		Amend 4			12/30/64				X	

11. SUBMITTED BY — DATE — R.J. GANGEMI

12. APPROVED BY — G. Coffey — George E. Coffey

AMC Form 1724R
26 Feb 1970

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